First- and second-order implementations for negatively comonotone problems

Abstract

We give two implementations of the anchored extra-proximal framework of Jiang and Yoon through the classical monotone graph transformation for negative comonotonicity. For a composite inclusion, a prescribed finite contraction meets the relative-error condition using only original operator values and backward resolvents. For every strict admissible margin, its accelerated first-order oracle bound has no accuracy logarithm. For a smooth equation with a Lipschitz Jacobian, we give a second-order implementation using original value–Jacobian queries and known linear systems. A finite prefix, independent of the requested accuracy, supplies a guaranteed lower line-search bracket. At fixed known model parameters and a supplied initial-distance bound, the total second-order query count has accuracy dependence $O(\epsilon^{-2/5}\log(2+1/\epsilon))$, in addition to the explicit prefix cost. Every trial and linear solve is counted. The graph transformation, acceleration energy, second-order exponent and search architecture are inherited; the implementation estimates connect them to the stated original oracles.

Keywords: comonotone inclusion; anchored extra-proximal method; oracle complexity; relative error

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The two oracle models are distinct. Sections 1–4 treat the composite first-order problem. Section 5 adds Jacobian access and a supplied distance bound for the smooth equation case. The first-order bound has no accuracy logarithm; the second-order bound retains its line-search logarithm.

1 Model and relation to existing results

Let $F:\mathbb{R}^d\to\mathbb{R}^d$ be globally L-Lipschitz, where $L>0$ is known, and let $A:\mathbb{R}^d\rightrightarrows\mathbb{R}^d$ be maximally monotone. Write $S=F+A$ and assume $\operatorname{zer} S\neq\varnothing$. A known signed parameter ρ satisfies

$$-1/L<\rho\le 0,\qquad \langle v-w,z-y\rangle\ge\rho\,\|v-w\|^2\quad((z,v),(y,w)\in\operatorname{gph} S).\tag{1}$$

The hypothesis concerns every two graph pairs of the sum. It does not require monotonicity of F alone. The available oracles are point evaluations of F and resolvents $J_{\gamma A}=(I+\gamma A)^{-1}$; all such calls, including inner work and the returned certificate, are counted.

Alacaoglu, Kim and Wright already obtain accelerated outer residual bounds throughout this negative range by inexact Halpern iteration [3]. Their implemented bound has

an accuracy-dependent logarithm arising from increasing inner work. Pethick, Mavrothalassitis and Cevher remove that logarithm for an unaccelerated squared-residual bound under the weaker weak-Minty condition. They ask about acceleration in the remaining lower half of the negative range [4, Section 10]. The explicit FEG and composite-FEG bounds cover $\rho > -1/(2L)$ [6, 7].

The recent anchored extra-proximal (AEP) framework supplies an accelerated bound for monotone inclusions under a predictor-relative error test [1, Theorem 3.2]. We use its energy argument after the standard graph transformation associated with comonotonicity [2]. The additional estimate below shows that a specified, finite inner contraction meets that error test using the original F and J_A oracles. No exact resolvent of S or of the transformed graph is implemented. The resulting original tangent certificate has $O(1/k)$ norm, with a constant number of oracle calls per outer step for each fixed margin.

2 Implemented iteration and residual bound

Define

$$r = -\rho, \quad \Delta = 1 - rL \in (0, 1], \quad \gamma = \frac{1 - \Delta/2}{L}, \quad a = \frac{\Delta}{2L}, \quad m \geq \left\lceil \frac{2}{\Delta} \log \frac{12}{\Delta} \right\rceil, \tag{2}$$

where m is an integer chosen once and log is natural. In particular, $\gamma = r + a$ and $\gamma L < 1$.

Choose any anchor $u_0 \in \mathbb{R}^d$ and put $v_0 = 0$. The placeholder $z_0 = u_0$ need not be in $\operatorname{dom} A$ and is not asserted to form a graph pair with v_0. For $k = 0, 1, \ldots$, set

$$t_k = \frac{k}{k+1}, \qquad b_k = (1 - t_k)u_0 + t_k u_k, \qquad w_{k,0} = b_k - \gamma t_k v_k, \tag{3}$$

perform the fixed inner loop

$$w_{k,j} = J_{\gamma A}\big(b_k - \gamma F(w_{k,j-1})\big), \qquad j = 1, \ldots, m, \tag{4}$$

and define

$$\begin{aligned} z_{k+1} &= w_{k,m}, \\ v_{k+1} &= \frac{b_k - w_{k,m}}{\gamma} + F(w_{k,m}) - F(w_{k,m-1}), \\ u_{k+1} &= z_{k+1} + r v_{k+1}. \end{aligned} \tag{5}$$

Cache the values $F(w_{k,j})$. The reported point and residual are (z_k, v_k) for $k \geq 1$.

Theorem 1. *Under* (1)*, the iteration* (2)–(5) *is well-defined and* $v_k \in S(z_k)$ *for every* $k \geq 1$*. For every* $p \in \operatorname{zer} S$*, with* $R = \|u_0 - p\|$*,*

$$\operatorname{dist}(0, S(z_k)) \leq \|v_k\| \leq \frac{4LR}{\Delta k}, \qquad k \geq 1. \tag{6}$$

One outer update uses at most $m + 1$ *evaluations of* F *and* m *resolvent calls. For any* $\epsilon > 0$*, stopping at the first computed pair with* $\|v_k\| \leq \epsilon$ *therefore uses at most*

$$(2m + 1) \max\left\{1, \left\lceil \frac{4LR}{\Delta \epsilon} \right\rceil\right\} \tag{7}$$

total primitive calls. The stopping rule does not require R. No boundedness assumption on the iterates is needed.

At $\rho = -3/(4L)$, one may take $m = 32$: the bound is $16LR/k$ with at most 65 calls per outer update. The constants are not optimized. At the smallest allowed m, the bound (7) is at most

$$\left(\frac{4}{\Delta}\log\frac{12}{\Delta}+3\right)\left(1+\frac{4LR}{\Delta\epsilon}\right). \tag{8}$$

The logarithm depends only on the known model margin, and no uniformity as $\Delta \downarrow 0$ is claimed. The tolerance in this statement is a residual *norm*, not a squared norm.

3 A fixed number of inner steps meets the error test

For each outer update define the predictor and the two errors

$$h_k = b_k - at_kv_k, \qquad \delta_k = u_{k+1} - h_k, \qquad e_k = u_{k+1} + av_{k+1} - b_k. \tag{9}$$

Lemma 1. *Every update in* (4) *is well-defined, its output satisfies* $v_{k+1} \in S(z_{k+1})$*, and*

$$\|e_k\| \le \tfrac{1}{2}\|\delta_k\| \qquad (k \ge 0). \tag{10}$$

The estimate holds uniformly over the input state and does not use its boundedness.

Proof. The resolvent of a maximally monotone operator is everywhere defined and non-expansive. Thus all finite inner loops are well-defined. The last resolvent step gives

$$c_{k+1} := \frac{b_k - w_{k,m}}{\gamma} - F(w_{k,m-1}) \in A(w_{k,m}),$$

and (5) gives $v_{k+1} = F(w_{k,m}) + c_{k+1} \in S(z_{k+1})$. No initial graph selection is used.

Fix an outer index and suppress it. Put $q = \gamma L = 1 - \Delta/2 \in [1/2, 1)$. The map

$$C_b(w) = J_{\gamma A}(b - \gamma F(w))$$

is a q-contraction. For analysis only, let $\bar{z}$ be its unique fixed point and put

$$\bar{v} = (b - \bar{z})/\gamma \in S(\bar{z}), \qquad \bar{u} = \bar{z} + r\bar{v}.$$

Then $b = \bar{z} + \gamma\bar{v} = \bar{u} + a\bar{v}$. The initial inner point and the predictor consequently obey the exact relation

$$\bar{z} - w_0 = \gamma(tv_k - \bar{v}) = \frac{\gamma}{a}(\bar{u} - h). \tag{11}$$

This comparison uses the actual prescribed starting point. Let $D = \|\bar{u} - h\|$. Contraction gives

$$\|w_j - \bar{z}\| \le q^j\frac{\gamma}{a}D \qquad (j \ge 0). \tag{12}$$

Neither $\bar{z}$ nor the iteration to its limit is computed by the algorithm.

Write (z, v, u) for the output of this update. Since $\gamma = r + a$, its error in (9) satisfies

$$e = z + \gamma v - b = \gamma\big(F(w_m) - F(w_{m-1})\big). \tag{13}$$

Using (12) at m and $m-1$ therefore yields

$$\|e\| \le (1+q)\frac{\gamma}{a} q^m D. \tag{14}$$

Also $v - \bar{v} = (e - z + \bar{z})/\gamma$, whence

$$\begin{aligned} u - \bar{u} &= \frac{a}{\gamma}(z - \bar{z}) + \frac{r}{\gamma} e, \\ \|u - \bar{u}\| &\le \left(1 + \frac{r}{a}(1+q)\right) q^m D. \end{aligned} \tag{15}$$

Both coefficients in (14) and (15) are at most $4/\Delta$. Indeed, $\gamma/a = 2q/\Delta$, and the first coefficient minus the second is q.

The choice of m gives

$$q^m \le \exp(-m\Delta/2) \le \Delta/12. \tag{16}$$

Equations (14)–(16) imply

$$\|e\| \le D/3, \qquad \|u - \bar{u}\| \le D/3, \qquad \|u - h\| \ge 2D/3.$$

Hence $\|e\| \le \|u - h\|/2$, as required. If $D = 0$, the same estimates make both errors zero; no cancellation of a possibly vanishing displacement is used. The proof includes $\Delta = 1$, equivalently $\rho = 0$. ∎

4 The AEP energy and original oracle bound

The graph relation in (1) gives, for all computed pairs with positive indices,

$$\langle v_i - v_j, u_i - u_j \rangle = \langle v_i - v_j, z_i - z_j \rangle + r \|v_i - v_j\|^2 \ge 0. \tag{17}$$

This is the classical monotone graph transformation [2]. The following potential is the constant-step AEP potential [1, Lemma A.2]:

$$U_k = \frac{a^2 k^2}{2} \|v_k\|^2 + ak\langle v_k, u_k - u_0 \rangle, \qquad k \ge 1. \tag{18}$$

We verify its descent on the actual computed graph pairs, including the first step from the arbitrary anchor.

Lemma 2. *For $k \ge 1$,*

$$\begin{aligned} U_k - U_{k+1} &= \frac{(k+1)^2}{2}\big(\|\delta_k\|^2 - \|e_k\|^2\big) \\ &\quad + k(k+1)a\left(\langle v_{k+1} - v_k, z_{k+1} - z_k \rangle + r\|v_{k+1} - v_k\|^2\right) \\ &\ge \frac{3}{8}(k+1)^2 \|\delta_k\|^2 . \end{aligned} \tag{19}$$

Moreover, $U_1 \le 0$.

Proof. The actual definitions give

$$\delta_k = e_k - av_{k+1} + a\frac{k}{k+1}v_k, \qquad u_{k+1} - u_k = \frac{u_0 - u_k}{k+1} - av_{k+1} + e_k.$$

Substitution of these two identities into (18), followed by expansion of the squared norm, gives

$$U_k - U_{k+1} = \frac{(k+1)^2}{2}\left(\|\delta_k\|^2 - \|e_k\|^2\right) + k(k+1)a\langle v_{k+1} - v_k, u_{k+1} - u_k\rangle.$$

Using $u_j = z_j + rv_j$ gives the identity in (19). Its inequality follows from (1) and Lemma 1.

At $k = 0$, $t_0 = 0$ and $h_0 = b_0 = u_0$. Thus $\delta_0 = u_1 - u_0$ and $e_0 = \delta_0 + av_1$. Directly,

$$U_1 = \tfrac{1}{2}\|e_0\|^2 - \tfrac{1}{2}\|\delta_0\|^2 \le -\tfrac{3}{8}\|\delta_0\|^2 \le 0.$$

This calculation invokes no graph pair at index zero. It is the same first-energy calculation used in the AEP argument, valid here for the nongraph placeholder initialization. ∎

Proof of Theorem 1. Well-definedness and graph membership follow from Lemma 1. Lemma 2 gives $U_k \le U_1 \le 0$. For an original zero p, (1) and (5) imply

$$\langle v_k, u_k - p\rangle = \langle v_k, z_k - p\rangle + r\|v_k\|^2 \ge 0.$$

Consequently,

$$0 \ge U_k \ge \frac{a^2k^2}{2}\|v_k\|^2 - akR\|v_k\|. \tag{20}$$

If $\|v_k\| > 0$, division by $ak\|v_k\|$ gives $\|v_k\| \le 2R/(ak)$; if the residual is zero the bound already holds. Inserting $a = \Delta/(2L)$ proves (6). This also covers $R = 0$, without division by R or a boundedness premise.

Each update first evaluates $F(w_{k,0})$. Each of the m inner steps then uses one resolvent and one evaluation at its new point. The last two evaluations already supply (5), so there are at most $m + 1$ evaluations of F and m resolvent calls. The initialization and starting-point formula require only arithmetic. No certificate-conversion call is needed because $v_k \in S(z_k)$ is already known.

For $\epsilon > 0$, put $K_\epsilon = \max\{1, \lceil 4LR/(\Delta\epsilon)\rceil\}$. Bound (6) ensures termination by K_ϵ; at least one computed pair is used because index zero is only a placeholder. Multiplication by the per-update count proves (7). The ceiling inequality gives (8). For a prescribed oracle budget $N \ge 2m + 1$, scheduling $K = \lfloor N/(2m+1)\rfloor$ complete updates fits that budget and returns the bound (6) at K. This is a guaranteed scheduled count, independent of accidental evaluation coincidences. ∎

5 A second-order implementation with a finite prefix

We now consider a smooth equation, with no backward operator. Let $F : \mathbb{R}^d \to \mathbb{R}^d$ be continuously differentiable in a finite dimension, globally L-Lipschitz, and have an

M-Lipschitz Jacobian in operator norm. The constants $L, M > 0$ and $r > 0$ are known, and

$$0 < rL < 1, \qquad \langle F(x) - F(y), x - y \rangle \geq -r \left\| F(x) - F(y) \right\|^2 \quad (x, y \in \mathbb{R}^d). \tag{21}$$

Assume $\operatorname{zer} F \neq \varnothing$. The input consists of x_0, $\epsilon > 0$, and a known $R > 0$ such that $\operatorname{dist}(x_0, \operatorname{zer} F) \leq R$. An original oracle call at x returns $(F(x), DF(x))$. A value-only call may replace it when the derivative is unused, but is counted conservatively as one original call below. Each solve of a specified, known $d \times d$ linear system is counted separately. The model uses exact real arithmetic; these counts do not measure bit complexity or practical runtime. In particular, a full Jacobian returns d^2 entries, and a linear solve has dimension-dependent arithmetic cost.

Put

$$\begin{aligned} &\Delta = 1 - rL, \qquad a_0 = \frac{\Delta}{2L}, \qquad \gamma_0 = r + a_0, \qquad m = \left\lceil \frac{2}{\Delta} \log \frac{12}{\Delta} \right\rceil, \\ &\overline{R} = (1 + rL)R, \qquad T_0 = \max\left\{ \frac{1}{L}, 112 M r^3 L \overline{R} \right\}, \qquad K_0 = \left\lceil \frac{T_0}{a_0} \right\rceil. \end{aligned} \tag{22}$$

The prefix length K_0 is independent of ϵ. The scalar A below denotes accumulated step size, not the set-valued operator of Sections 1–4.

Initialization and prefix. Query $(F(x_0), DF(x_0))$ and return x_0 if $\|F(x_0)\| \leq \epsilon$. Otherwise set

$$g = F(x_0), \quad x = x_0, \quad u = x_0 + rg, \quad u_{\mathrm{a}} = u, \quad A = 0.$$

The anchor u_{a} stays fixed. Perform at most K_0 updates, each with $a = a_0$, $\gamma = \gamma_0$:

$$\begin{aligned} &t = \frac{A}{A + a}, && b = (1 - t)u_{\mathrm{a}} + tu, \quad w_0 = b - \gamma t g, \\ &w_j = b - \gamma F(w_{j-1}) \quad (1 \leq j \leq m), \\ &x^+ = w_m, && g^+ = F(x^+), && u^+ = x^+ + r g^+. \end{aligned} \tag{23}$$

Replace (x, g, u, A) by $(x^+, g^+, u^+, A + a)$ and return x whenever $\|g\| \leq \epsilon$. All evaluations are original calls, including $F(w_m)$. Each completed prefix update uses at most $m + 1$ such calls and no linear solve.

A Newton trial. At a post-prefix state, for any $a \geq r$ define

$$\begin{aligned} &\gamma = a + r, \qquad t = \frac{A}{A + a}, \qquad b = (1 - t)u_{\mathrm{a}} + tu, \qquad w = b - \gamma t g, \\ &(I + \gamma DF(w))x^+ = b - \gamma F(w) + \gamma DF(w) w, \\ &d = x^+ - w, \qquad g^+ = F(x^+), \qquad u^+ = x^+ + r g^+, \qquad \phi(a) = M\gamma \left\| d \right\|. \end{aligned} \tag{24}$$

Query the jet at w, solve the displayed known linear system, and query the original jet at x^+. A trial therefore uses at most two original calls and one linear solve. The derivative at x^+ need not be used. Rejected trial states are discarded.

Search and update. First evaluate the trial

$$U_k = \max\left\{r, \frac{2A\,\|g\|}{\epsilon}, \sqrt{\frac{5}{4M\epsilon}}\right\}. \tag{25}$$

If $\phi(U_k) \le 1/2$, return that trial's x^+. Otherwise evaluate $a = r$. We prove below that $\phi(r) \le 1/4$. Accept it if equality holds. If $\phi(r) < 1/4$, bisect $[r, U_k]$ arithmetically: accept a midpoint when

$$\frac{1}{4} \le \phi(a) \le \frac{1}{2}; \tag{26}$$

replace the lower endpoint if $\phi(a) < 1/4$, and the upper endpoint if $\phi(a) > 1/2$. No monotonicity of ϕ is required. At acceptance replace (x, g, u, A) by $(x^+, g^+, u^+, A + a)$. Return if $\|g\| \le \epsilon$, and otherwise repeat the search.

Theorem 2. *Under the hypotheses of this section, the algorithm is well-defined and returns an originally queried point x with $\|F(x)\| \le \epsilon$. Define*

$$\begin{gathered} N = \left\lceil \left(\frac{512M\overline{R}^2}{\epsilon}\right)^{2/5} \right\rceil, \qquad U = \max\left\{r, \frac{4\overline{R}}{\epsilon}, \sqrt{\frac{5}{4M\epsilon}}\right\}, \qquad \Gamma = U + r, \\ Q = 11L\overline{R}, \qquad C_P = L + 5\Gamma LM\overline{R}, \qquad C_q = 7L^2\overline{R}, \\ D = Q(1 + \Gamma C_P) + \Gamma C_q, \qquad H = M\Gamma(Q + D), \\ B = 3 + \left\lceil \log_2\big(\max\{1, 4UH\}\big) \right\rceil. \end{gathered} \tag{27}$$

The total counts are at most

$$1 + K_0(m+1) + 2NB \quad \textit{original calls}, \qquad NB \quad \textit{known linear solves}. \tag{28}$$

They include initialization, the prefix, both endpoints, every rejected midpoint and each final output query. If $j \ge 1$ Newton updates have been accepted without earlier return, their actual residual satisfies

$$\|g\| \le \frac{512M\overline{R}^2}{j^{5/2}}. \tag{29}$$

At fixed L, M, r, R with $\Delta > 0$, the accuracy dependence in (28), as $\epsilon \downarrow 0$, is $O(\epsilon^{-2/5}\log(2+1/\epsilon))$ in addition to the explicit prefix. No joint-parameter optimality is asserted.

The 2/5 exponent, relative-error energy and bracketed-search architecture are those of second-order AEP [1, Section 4]. The additional work is an implementation using original jets throughout: the finite prefix supplies a usable positive lower bracket, and the direct trial (24) supplies a true original graph certificate. It does not evaluate the nonlinear inverse of $x \mapsto x + rF(x)$.

6 Proof of the second-order bound

For every accepted step use the actual transformed predictor, displacement and error

$$h = b - atg, \qquad \zeta = u^+ - h, \qquad e = u^+ + ag^+ - b = x^+ + (a+r)g^+ - b. \tag{30}$$

The prefix uses $a = a_0$. Throughout this section fix a zero p with $\|x_0 - p\| \le R$; such a zero exists since the nonempty zero set is closed in finite dimension. It satisfies $\|u_{\mathrm{a}} - p\| \le \overline{R}$.

Lemma 3. *Every prefix update satisfies $\|e\| \le \|\zeta\|/2$. Every Newton trial is well-defined for $a \ge r$. If $\phi(a) \le 1/2$, it satisfies*

$$\|e\| \le \frac{1}{4}\|d\|, \qquad \|\zeta\| \ge \frac{3}{8}\|d\|, \qquad \|e\| \le \frac{2}{3}\|\zeta\|. \tag{31}$$

These estimates do not assume bounded states.

Proof. During the prefix, $A = ka_0$ and (23) is (3)–(5) with the backward operator equal to zero. The initial stored $g = F(x_0)$ is irrelevant to the first inner loop because $t = 0$. Subsequent outputs satisfy $g = F(x)$ exactly. Lemma 1 therefore applies, including the initial step and zero displacement.

For a Newton trial, differentiating (21) along a direction v gives

$$\langle DF(w)v, v\rangle \ge -r\,\|DF(w)v\|^2.$$

Thus, for $\gamma = a + r \ge 2r$,

$$\|(I + \gamma DF(w))v\|^2 \ge \|v\|^2 + \gamma(\gamma - 2r)\,\|DF(w)v\|^2 \ge \|v\|^2. \tag{32}$$

The square matrix is invertible in finite dimension and its inverse norm is at most one. This is the usual comonotone-resolvent bound [2], here verified directly for the implemented matrix.

The known model equation gives

$$e = \gamma\big(F(x^+) - F(w) - DF(w)d\big), \qquad \|e\| \le \frac{M\gamma}{2}\|d\|^2 = \frac{\phi(a)}{2}\|d\|.$$

The exact relations $e = d + \gamma(g^+ - tg)$ and

$$\zeta = d + r(g^+ - tg) = \frac{a}{\gamma}d + \frac{r}{\gamma}e$$

yield, when $\phi(a) \le 1/2$,

$$\|\zeta\| \ge \frac{a - r/4}{a + r}\|d\| \ge \frac{3}{8}\|d\|.$$

This proves (31), including $d = 0$ without division by a vanishing norm. ∎

Lemma 4. *Every finite sequence of accepted updates, in either phase, satisfies*

$$V(A,u,g) := \frac{A^2}{2}\|g\|^2 + A\langle g, u-u_{\mathrm{a}}\rangle \le 0,$$
$$\|g\| \le \frac{2\overline{R}}{A} \quad (A>0), \qquad \sum \frac{A_{\mathrm{next}}^2}{a^2}\|\zeta\|^2 \le \frac{9}{5}\overline{R}^2, \qquad \|u-p\| \le \left(1+\frac{2}{\sqrt{5}}\right)\overline{R} < 2\overline{R}. \tag{33}$$

The sum includes the prefix. No boundedness hypothesis is used.

Proof. The actual pairs obey $u = x + rF(x)$, so (21) implies

$$\langle g^+ - g, u^+ - u\rangle \ge 0, \qquad \langle g, u-p\rangle \ge 0.$$

The variable-step AEP identity [1, Lemma A.2], written for these actual pairs, is

$$\begin{aligned} V(A,u,g) - V(A+a,u^+,g^+) = {} & \frac{(A+a)^2}{2a^2}\left(\|\zeta\|^2 - \|e\|^2\right) \\ & + \frac{A(A+a)}{a}\langle g^+ - g, u^+ - u\rangle. \end{aligned} \tag{34}$$

It follows by expanding the potential with

$$(A+a)g^+ - Ag = \frac{A+a}{a}(e-\zeta), \qquad u^+ - u = -\frac{a}{A+a}(u - u_{\mathrm{a}} + Ag) + \zeta.$$

The identity holds also at $A=0$, when the second term vanishes. Lemma 3 gives decrease at least $5(A+a)^2\|\zeta\|^2/(18a^2)$.

On the other hand, comparison with p gives

$$V(A,u,g) \ge \frac{1}{2}\|Ag + p - u_{\mathrm{a}}\|^2 - \frac{1}{2}\|p-u_{\mathrm{a}}\|^2 \ge -\frac{1}{2}\overline{R}^2.$$

Starting from $V=0$, telescoping proves the energy budget. The residual estimate follows from

$$V \ge \frac{A^2}{2}\|g\|^2 - A\overline{R}\,\|g\|,$$

treating $g=0$ separately.

It remains to bound the actual state. Since $u^+ + ag^+ = b + e$ and $\langle g^+, u^+ - p\rangle \ge 0$,

$$\left\|u^+ - p\right\| \le \|b-p\| + \|e\| \le \frac{A}{A+a}\|u-p\| + \frac{a}{A+a}\overline{R} + \|e\|.$$

The already proved budget gives

$$\frac{A+a}{a}\|e\| \le \frac{2}{3}\frac{A+a}{a}\|\zeta\| \le \frac{2}{\sqrt{5}}\overline{R}.$$

Induction from $\|u_{\mathrm{a}} - p\| \le \overline{R}$ gives the last bound in (33). In particular, $\|u - u_{\mathrm{a}}\| \le 3\overline{R}$ at every stored state. ∎

Lemma 5. *At every reached Newton state, the lower trial satisfies $\phi(r) \le 1/4$. If the upper trial passes $\phi(U_k) \le 1/2$, its actual output has $\|F(x^+)\| \le \epsilon$.*

Proof. After the prefix, $A = K_0 a_0 \ge T_0 \ge 1/L > r$, and A increases afterwards. From the explicit predictor and $x = u - rg$,

$$w - x = \frac{a}{A+a}\big(u_{\mathrm{a}} - u + (r-A)g\big). \tag{35}$$

At $a = r$, Lemma 4 gives

$$\|w - x\| \le \frac{5r\overline{R}}{A}, \qquad \|F(w) - tg\| \le \frac{5rL\overline{R}}{A} + \frac{2r\overline{R}}{A^2} \le \frac{7rL\overline{R}}{A}.$$

Equation (24) implies $d = -\gamma(I + \gamma DF(w))^{-1}(F(w) - tg)$. At $\gamma = 2r$, (32) yields

$$\|d\| \le \frac{14r^2L\overline{R}}{A}, \qquad \phi(r) \le \frac{28Mr^3L\overline{R}}{A} \le \frac{1}{4}.$$

This is the purpose of the finite prefix.

For any trial with $\phi(a) \le 1/2$, the exact identity $g^+ = tg - d/\gamma + e/\gamma$ and Lemma 3 give

$$\big\|g^+\big\| \le \frac{A\,\|g\|}{a} + \frac{5}{8M\gamma^2} \le \frac{A\,\|g\|}{a} + \frac{5}{8Ma^2}. \tag{36}$$

At $a = U_k$, each term is at most $\epsilon/2$. This certifies the original value already queried at x^+; it requires no subsequent accepted step. Also $A\,\|g\| \le 2\overline{R}$ implies $U_k \le U$ in (27). ∎

Lemma 6. *Every Newton search terminates after at most B trials, where B is given in* (27).

Proof. Fix a current state and $a \in [r, U]$. Write

$$P(a) = I + (a+r)DF(w(a)), \qquad c(a) = F(w(a)) - t(a)g, \qquad d(a) = -(a+r)P(a)^{-1}c(a).$$

Equations (35) and (33) show

$$\|w - x\| \le 5\overline{R}, \qquad \|c(a)\| \le 7L\overline{R} \le Q.$$

Differentiating just the scalar rational coefficients gives

$$w'(a) = \frac{A}{(A+a)^2}\big(u_{\mathrm{a}} - u - (A-r)g\big), \qquad \|w'(a)\| \le \frac{5\overline{R}}{A} \le 5L\overline{R}.$$

As $|t'(a)| \le 1/A$, the map c is C_q-Lipschitz. The global Lipschitz bound on F implies $\|DF(w)\| \le L$, so the Lipschitz bound on DF makes P C_P-Lipschitz. By (32), its inverse has norm at most one. The identity for the difference of two inverses therefore makes P^{-1} C_P-Lipschitz as well. Consequently

$$\|d(a)\| \le \Gamma Q, \qquad \mathrm{Lip}(d; [r, U]) \le Q(1 + \Gamma C_P) + \Gamma C_q = D,$$

and $\mathrm{Lip}(\phi;[r,U]) \le M\Gamma(Q+D) = H$. These are Lipschitz estimates; differentiability of DF is not required.

If the upper trial does not return and the lower endpoint is not accepted, the bracket satisfies $\phi(\mathrm{lo}) < 1/4$ and $\phi(\mathrm{hi}) > 1/2$. Every unsuccessful midpoint preserves these strict inequalities and halves the interval length. Such an interval must have length greater than $1/(4H)$. Starting with length at most U, this is impossible after $\lceil \log_2(\max\{1, 4UH\}) \rceil$ unsuccessful halvings. Adding both endpoint trials and one conservative midpoint allowance gives B. If $U_k = r$, upper failure would contradict the proved lower bound at that same point, so no degenerate failing bracket arises. ∎

Proof of Theorem 2. The prefix is finite and well-defined. For any reached Newton state, Lemma 3 makes all trial linear solves well-defined. Lemmas 5 and 6 then guarantee either a certified return or a finite accepted step. Its error estimate preserves Lemma 4. This induction justifies use of the state bounds at every later search, with no circular boundedness premise.

At an accepted Newton step, (26) and $a + r \le 2a$ give

$$\|d\| \ge \frac{1}{4M(a+r)} \ge \frac{1}{8Ma}, \qquad \|\zeta\| \ge \frac{3}{64Ma}.$$

Let A_i be the full accumulated step after the ith Newton acceptance, and a_i its step. The prefix mass is positive. The full energy budget implies

$$\sum_{i=1}^{j} \frac{A_i^2}{a_i^4} \le C, \qquad C = \frac{4096}{5} M^2 \overline{R}^2. \tag{37}$$

For $s_i = a_i/\sqrt{A_i}$, Hölder's inequality gives

$$j \le \left(\sum_{i=1}^{j} s_i \right)^{4/5} C^{1/5}.$$

Moreover,

$$\frac{a_i}{\sqrt{A_i}} \le 2(\sqrt{A_i} - \sqrt{A_{i-1}}), \qquad \sum_{i=1}^{j} s_i \le 2\sqrt{A_j}.$$

Thus $A_j \ge j^{5/2}/(4\sqrt{C})$, and the actual residual satisfies

$$\|g\| \le \frac{512}{\sqrt{5}} \frac{M\overline{R}^2}{j^{5/2}} \le \frac{512 M \overline{R}^2}{j^{5/2}}.$$

The stated coarse constant is retained in the stopping budget. An accepted Nth Newton step must return by its residual test, while an earlier upper-trial return is certified by (36). Hence there are at most N Newton searches, each with at most B trials. Adding the initial query and all prefix calls gives (28).

All equalities at the acceptance thresholds and at $\|g\| = \epsilon$ use the displayed closed branches. If a trial has $d = 0$, then $e = \zeta = 0$ and $\phi = 0$; it either returns at the upper

endpoint or moves the lower endpoint, without division by that norm. The matrix bound includes $a = r$. If $R = 0$, closedness of $\operatorname{zer} F$ forces $F(x_0) = 0$, so the single initial query returns.

Finally, as $\epsilon \downarrow 0$ with the supplied model parameters fixed, $U = O(\epsilon^{-1})$, $D = O(\epsilon^{-2})$ and $H = O(\epsilon^{-3})$. Thus $B = O(\log(2 + 1/\epsilon))$ and $N = O(\epsilon^{-2/5})$. The explicit count, rather than this asymptotic notation alone, applies to every positive accuracy. ∎

7 Attribution and scope

The graph transformation used here is

$$\operatorname{gph} B = \{(z + rv, v) : (z, v) \in \operatorname{gph} S\}.$$

Its monotonicity is (17), equivalently the classical relation between S and $S^{-1} - \rho I$ [2]. The analysis-only contraction point in Lemma 1 solves $b = \bar{u} + a\bar{v}$ with $(\bar{u}, \bar{v}) \in \operatorname{gph} B$. This explains the use of the monotone AEP energy. It does not grant an additional implemented oracle.

The inner approximation by repeated F evaluations and A resolvents is established in RAPP [5] and in the relative-error analysis of [4]. The latter already obtains inner work independent of the target accuracy for its own halfspace-corrected, unaccelerated iteration. Monotone envelope transformations also appear in the FEG ancestry [6, Appendix A.1]. AEP supplies the predictor-relative error condition and the accelerated energy bound [1]. The first-order implementation estimate is (11)–(16), which makes a fixed inner count satisfy that condition on the transformed graph while retaining an original graph certificate. The arbitrary-anchor first step is checked directly above; AEP's stated algorithm asks for an initial graph element.

The substantive comparison is in $\rho \in (-1/L, -1/(2L)]$: the total bound (7) removes the accuracy-dependent logarithm of the full-range inexact-Halpern implementation. Accelerated outer convergence on that range was already established by [3]. For the upper half of the range and at $\rho = 0$, explicit accelerated methods were already available [1, 6, 7]. We make no claim of better numerical constants there, an optimal dependence on Δ, or a new abstract acceleration principle. The comparison with [3] uses its version-pinned Theorem 2.1 and Corollary 2.2; the current upload date does not date the original ICML 2024 result.

The proof requires all graph pairs in (1); weak Minty only against a zero does not supply the consecutive-pair inequality in (19). The model parameters L and ρ are known. No conclusion is asserted at $\rho = -1/L$, and no point-selection, noise or unknown-parameter guarantee is included. The first-order result is an original-oracle implementation with fixed inner work for each fixed admissible margin. The second-order result has the additional hypotheses and distinct oracle convention stated in Section 5.

For the second-order part, the inherited graph transformation would formally define

$$B(v) = F\big((I + rF)^{-1}v\big).$$

An exact value–Jacobian query to B at a prescribed point entails an unknown nonlinear inverse. It is not one original jet query or one known matrix solve. The direct base point in (24) instead satisfies

$$(w + rF(w)) - h = r(F(w) - tg),$$

which need not vanish. Thus (24) is not simply Algorithm 2 of [1] with renamed coordinates. Its relative-error control, the finite lower-bracket prefix, and the trial count make the inherited framework executable in the stated original oracle model. The affine inverse bound is classical [2]; the energy, 2/5 exponent and bracketed-search mechanism are AEP [1]. No new acceleration principle is asserted.

Higher-order nonmonotone results under an ordinary Minty condition have different hypotheses and residual criteria [8, 9]. Pairwise negative comonotonicity does not imply that condition: for a planar quarter-turn J, $F(x) = (-I + 2J)x$ has $L = \sqrt{5}$ and $r = 1/5$, while $\langle F(x), x\rangle = -\|x\|^2$. Its unique zero is not Minty. Theorem 2 is a restricted implementation result for a smooth equation with known L, M, r, R; it does not extend the composite first-order theorem to arbitrary backward operators at second order. The prefix can be large, and its cost is retained in (28). Neither joint-parameter optimality nor a matching lower bound for this precise class is claimed. The comparison with [8] uses its accepted manuscript; no exhaustive priority conclusion is drawn from the bounded source comparison.

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